SIEM Log Monitoring & Threat Detection

Detecting Windows failed-logon activity (Event ID 4625) with Wazuh SIEM

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GitHub repository: https://github.com/Charlie9548/SOC-Analyst-L1-Portfolio.git

Project type: SOC Analyst L1 — practical lab

Platform: Wazuh SIEM (Manager on Ubuntu Linux)

Monitored endpoint: Windows 10 Pro, agent name *mylaplab*

Evidence date: 28 October 2026

Executive Summary

This lab shows the first stage of a SOC authentication-monitoring workflow. A failed Windows sign-in was generated on a monitored endpoint, confirmed in the native Windows Security log with PowerShell, and then found as alerts in the Wazuh dashboard. The Wazuh Threat Hunting search returned **6 matching alerts** for Windows Event ID 4625 on agent *mylaplab*, with the rule description *Logon Failure - Unknown user or bad password* at rule level 5.

What this proves: the telemetry pipeline works end to end (endpoint, agent, manager, indexer, dashboard), and I can search and read the resulting alerts.

> **Scope note**
> Six failed-logon events are not, by themselves, proof of a brute-force attack. Source IP, target account, logon type and failure reason must be reviewed before any incident determination. Those fields have not yet been documented, and this report does not claim an attack.

Skills demonstrated

- Verifying a SIEM agent service and endpoint enrolment on Windows.
- Querying the Windows Security log with PowerShell (`Get-WinEvent`).
- Searching and reading alerts in Wazuh Threat Hunting.
- Separating observed evidence from assumptions, which is a core SOC habit.

1. Objective and Scope

The goal was to understand how a SIEM collects and presents endpoint security events, and how a SOC analyst starts an authentication-failure investigation.

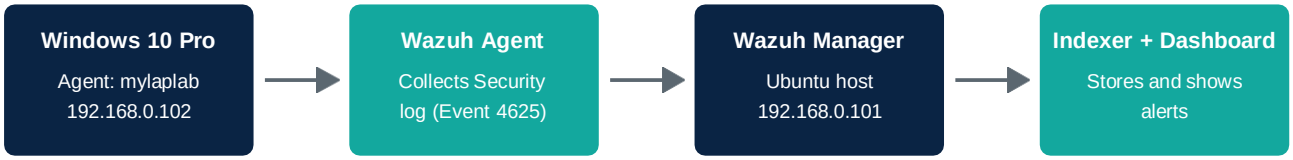
In scope

- Verify that the Windows 10 Wazuh Agent service is running.
- Verify Windows Security Event ID 4625 using PowerShell.
- Search Wazuh Threat Hunting for Event ID 4625 and review the alert description, timestamps and rule level.

Out of scope (not yet completed)

- A custom correlation rule for repeated failures.
- A confirmed brute-force classification.
- Full review of source IP, target username and failure reason for every event.
- A documented response or remediation action.

2. Lab Environment and Architecture



Analyst then runs a Threat Hunting search in the Dashboard: data.win.system.eventID: "4625"

Figure 1: Data flow from the Windows endpoint to the Wazuh dashboard.

Wazuh Manager / dashboard host: Ubuntu Linux, manager IP shown in lab: 192.168.0.101

Monitored endpoint: Windows 10 Pro, agent name mylaplab, dashboard IP shown: 192.168.0.102

Search used in Wazuh: `data.win.system.eventID: "4625"`

Verification tool: PowerShell, `Get-WinEvent` on the Security log, Event ID 4625

3. Background: Windows Event ID 4625

Definition: Event ID 4625 is written to the Windows Security log each time an account fails to log on. The message text is "An account failed to log on." It is one of the most common starting points for authentication investigations.

Fields an analyst reads first

- **TargetUserName** — the account that was tried. A non-existent name suggests guessing; a real admin name raises priority.
- **IpAddress / WorkstationName** — where the attempt came from. One source with many failures is a stronger signal than many sources with one failure each.
- **LogonType** — how the logon was attempted. Type 2 is interactive (keyboard), 3 is network (for example SMB), 7 is unlock, and 10 is RemoteInteractive (RDP).
- **Status / SubStatus** — the reason. For example 0xC000006D is a generic logon failure, 0xC000006A is a correct user with a wrong password, and 0xC0000064 is a user name that does not exist.

Why this matters: a failed logon at the keyboard (type 2) with a mistyped password is usually harmless. Repeated type 3 or type 10 failures from an external address deserve a closer look.

4. Methodology

Step 1 — Check the agent service

PowerShell was used to confirm the Wazuh agent service on the Windows endpoint:

```
Get-Service WazuhSvc
```

Result: the service was in the **Running** state and the dashboard listed the endpoint as *mylaplab*. **OBSERVED**

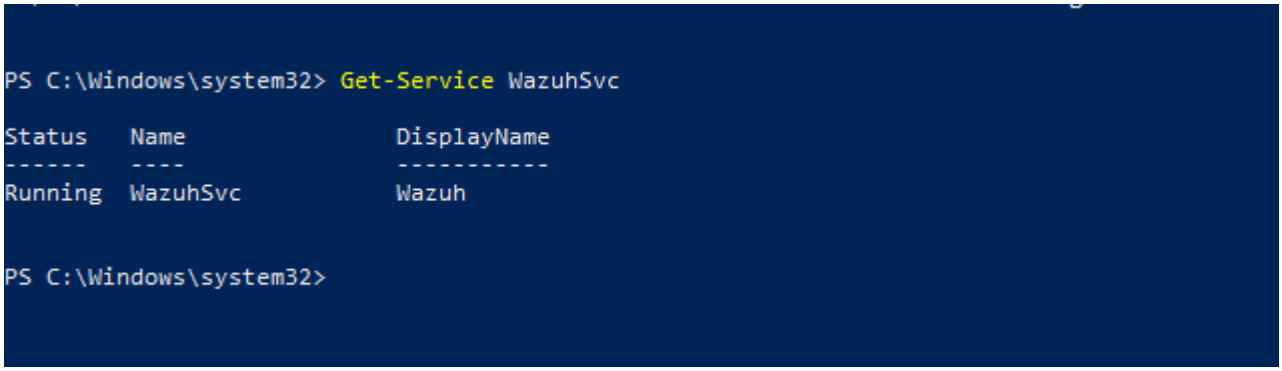


Figure 2: Wazuh agent service status and dashboard agent list.

Step 2 — Generate and verify failed-logon events

A failed sign-in was generated on the lab Windows endpoint, then the Security log was inspected:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 5
```

Result: the command returned Windows Security-Auditing events with Event ID 4625 and the message "An account failed to log on." **OBSERVED**

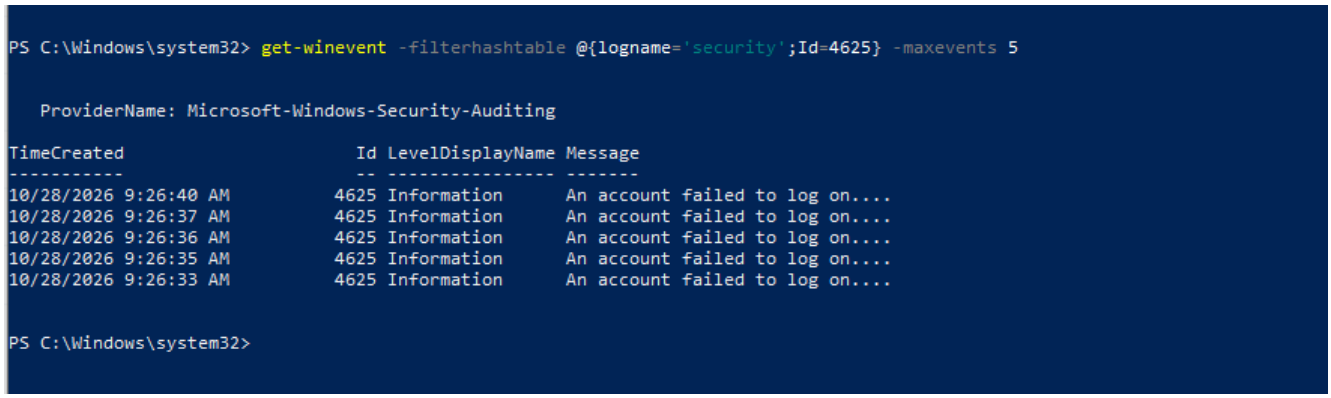


Figure 3: PowerShell output showing Event ID 4625.

Step 3 — Find the alerts in Wazuh

In **Threat Hunting > Events**, the following search was used:

```
data.win.system.eventID: "4625"
```

Result: 6 hits were shown in the Events table. **OBSERVED**

5. Findings

Matching events: 6 hits in the Wazuh Events table

Agent: mylaplab (Windows 10 Pro)

Rule description: Logon Failure - Unknown user or bad password

Rule level: 5

Event identifier: Windows Event ID 4625

Time window: 27 Oct 2026 to 28 Oct 2026 (range displayed in the screenshot)

If the pattern were confirmed as repeated guessing, it would map to **T1110 Brute Force**, for example T1110.001 (Password Guessing) or T1110.003 (Password Spraying). This is a candidate mapping for reference, not a finding from this lab.

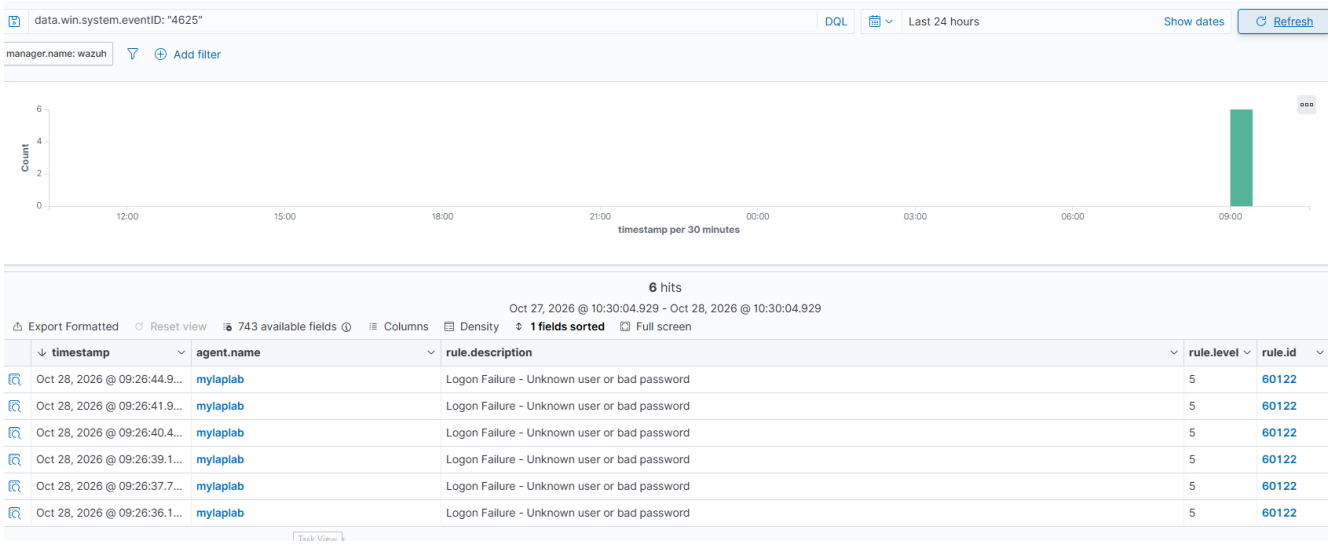


Figure 4: Wazuh Threat Hunting results for Event ID 4625 (6 hits).

Interpretation

Windows failed-logon events were present in the Security log, and six matching alerts were visible in Wazuh. The rule description points to a logon failure caused by an unknown user or a bad password. Wazuh places level 5 in its user-generated-errors range, which fits a low-to-medium severity single failure, not a confirmed incident.

> **Analyst caution**
>
> Do not label this as brute force yet. Inspect target username, source IP, logon type, failure status and substatus, timing, and any successful logon that follows the failures.

6. Triage Approach (how I would investigate next)

This section is a method, not a result. It shows how I would turn the six alerts into a decision.

Questions to answer

- Are all six failures for the same target account, or for different accounts?
- Do they share one source IP, or come from several?
- What logon type is it (keyboard, network, RDP)?
- Are the failures clustered in seconds or minutes, or spread out over hours?
- Is there a successful logon (Event ID 4624) right after the failures?

Leans benign

- One account, keyboard logon, a few failures, then success shortly after.
- Source is the local machine or a known admin workstation.
- Timing matches a user working hours.

Leans suspicious

- Many failures in a short window from one external or unknown source.
- Many different usernames from one source (a sign of password spraying).
- Failures followed by a success on the same account.
- Network or RDP logon types, or activity outside normal hours.

MITRE ATT&CK mapping (candidate only)

Optional: summarise 4625 events by account and source in PowerShell

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 50 |
ForEach-Object {
  $d = ([xml]$_.ToXml()).Event.EventData.Data
  [pscustomobject]@{
    Time       = $_.TimeCreated
    User       = ($d | ? Name -eq 'TargetUserName').'#text'
    SourceIP   = ($d | ? Name -eq 'IpAddress').'#text'
    LogonType  = ($d | ? Name -eq 'LogonType').'#text'
    SubStatus  = ($d | ? Name -eq 'SubStatus').'#text'
  }
} | Format-Table -AutoSize
```

7. Planned Detection Improvement (not yet implemented)

A single failed logon is normal noise. A SIEM becomes useful when it alerts on a **pattern**. The draft below is a Wazuh frequency rule that would raise severity when the same source produces several failures in a short time. It is an example to be tested in the lab, and it has not been run or validated.

```
<group name="windows,authentication_failures,">
  <rule id="100100" level="10" frequency="5" timeframe="120">
    <if_matched_sid>60122</if_matched_sid>
    <same_field>win.eventdata.ipAddress</same_field>
    <description>Repeated Windows logon failures from same source (lab rule)</description>
    <mitre><id>T1110</id></mitre>
  </rule>
</group>
```

- Place custom rules in `/var/ossec/etc/rules/local_rules.xml` on the manager and restart the manager after editing.
- Confirm the base rule ID (60122 is the usual ID for the unknown-user/bad-password rule) by opening the alert in your own dashboard.
- Test with `/var/ossec/bin/wazuh-logtest` using a sample 4625 event before relying on it.
- Document false positives: users who mistype passwords, service accounts with expired credentials, and scheduled tasks using old passwords.

8. Evidence Checklist

- Windows Wazuh Agent service running — **OBSERVED**
- Windows Security Event ID 4625 in PowerShell — **OBSERVED**
- Six matching Wazuh alerts — **OBSERVED**
- Expanded alert with user and source details — **NOT YET DOCUMENTED**
- Custom repeated-failure detection rule — **NOT IMPLEMENTED**
- Documented response or remediation action — **NOT IMPLEMENTED**

9. Next Steps

Phase 1 — Complete the investigation. Expand one alert and record target username, source IP, logon type and failure reason. Compare timestamps with successful logons.

Phase 2 — Build detection. Implement and test the repeated-failure rule in a controlled lab, then record how many alerts it produces and any false positives.

Phase 3 — Add response. Write a short playbook: verify the source, check the account, reset credentials if needed, and decide whether to block or escalate.

Phase 4 — Publish. Add redacted screenshots and the final README to GitHub.

10. Conclusion

This project gave me practical experience with Wazuh agent monitoring, Windows Security logs, Event ID 4625, and the first steps of a SIEM alert investigation. It is a foundation for authentication monitoring and can be extended into repeated-failure analysis and detection engineering.

> **Before publishing**
>
> Add your name, the repository URL and your own redacted screenshots. Do not include passwords, API keys, agent authentication keys, or other secrets. Consider masking internal IP addresses and usernames if the repository is public.

SIEM Log Monitoring & Threat Detection

Investigation Addendum — Windows Authentication Events

Author: Jeetu Rajput **Endpoint:** Windows 10 Pro / mylaplab (Agent ID 003) **Evidence date:** 28 October 2026

Purpose of this addendum

This addendum records the follow-up investigation performed after the original report. The original six-hit screenshot is retained as an initial observation. A later Last 7 days search for agent **mylaplab** and Event ID **4625** returned **8 failed-logon events**. A combined query for the account **sadnbox** and Event IDs 4624/4625 displayed **16 matching results** in the dashboard screenshot. Counts refer to their respective queries and time windows; they should not be treated as a single immutable total.

Evidence summary

Item	Observed evidence
Endpoint	mylaplab — Windows 10 Pro, Agent ID 003, agent IP 192.168.0.102
Failed logons	8 events (Event ID 4625) in the Last 7 days query; 6 clustered around 09:26 and 2 around 16:59 on 28 Oct 2026 (dashboard display time).
Wazuh rule	Rule ID 60122; rule level 5; “Logon Failure - Unknown user or bad password”.
Selected failed event	Target account sadnbox; Logon Type 2; Status 0xC000006D; SubStatus 0xC000006A (incorrect password); process svchost.exe; source 127.0.0.1.
Another selected failed event	Target account sadnbox; Logon Type 2; Status 0xC000006D; SubStatus 0xC000006A; process msedge.exe; source address unavailable (“-”).
Account verification	PowerShell Get-LocalUser showed sadnbox exists and Enabled=True.
Successful logon	A selected Event ID 4624 for sadnbox was observed, Logon Type 2, source 127.0.0.1; one selected event showed Elevated Token=Yes. Wazuh rule ID 60118, level 3.

Timeline interpretation

The failed-logon events are clustered: six occurred within roughly eight seconds around 09:26, and two occurred around 16:59. Successful logon events for sadnbox were also present. One selected successful event was timestamped around 16:59:08, before failed events around 16:59:39–16:59:41. Therefore, the evidence does **not** show that those failed attempts were followed immediately by a successful logon. Other successful events appear around the earlier cluster, but the provided evidence does not establish a direct causal relationship or same-session correlation.

Technical interpretation

Status **0xC000006D** is a general logon failure and substatus **0xC000006A** indicates an incorrect password. Logon Type 2 is an interactive logon type. Address **127.0.0.1** is the loopback/localhost address and does not identify an external source. A missing source address (“-”) cannot be used to identify a remote origin. The presence of svchost.exe or msedge.exe in the event is context for investigation, not proof that either process is malicious.

Investigation conclusion and next steps

Evidence-based conclusion

Multiple failed authentication events were observed on the Windows 10 endpoint, including a short cluster of six events and a later pair. Selected events identify the enabled local account sadnbox and an incorrect-password substatus. Successful logon events for the same account were also observed. The available evidence is consistent with repeated incorrect-password attempts or application/local authentication activity, but it does not establish brute force, password spraying, or account compromise.

What has been verified

• Wazuh manager and Windows endpoint agent were available in the dashboard; agent mylaplab was Active at the time of verification.

• Windows Security Event ID 4625 was queried in Wazuh Threat Hunting.

• Eight matching failed-logon events were visible for mylaplab in the later Last 7 days query.

• Selected failed events were expanded and their target account, logon type, failure status/substatus, process and source-address fields reviewed.

• The local account sadnbox was verified as present and enabled with Get-LocalUser.

• Selected Event ID 4624 success records for sadnbox were reviewed; no direct attack linkage was established.

Remaining limitations

• The combined 16-result screenshot shows matching 4624/4625 events; all rows have not been individually expanded to validate every username.

• The source address is localhost for some events and unavailable for another selected failed event; no external source was established.

• A custom repeated-failure detection rule has not been implemented or validated.

• No containment, credential reset, or remediation action was performed as part of this lab.

Recommended next actions

• If continuing the lab, export the matching 4624/4625 events and verify usernames and timestamps row by row.

• Inspect the Windows event context and relevant application/service activity around the clustered failures; avoid assuming the process name alone explains the event.

• Only test a correlation rule after confirming the correct event fields and validating it with wazuh-logtest; the earlier draft rule remains untested.

• Update the GitHub README/report link to point to this v3 report after uploading it. Do not publish credentials, agent keys, or other secrets.

Portfolio statement: This is a lab investigation based on observed telemetry. No confirmed malicious activity or compromise is claimed. Rule level 5 is described here as a low-severity alert, not a confirmed incident.